NEWTON SCHOOL

PHYSICS HONORS | ANSWER KEY

Sound & Waves Homework — icp13 · Solutions

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***For teacher reference** — full step-by-step solutions for all 14 problems.*

SOLUTIONS

Problem 1

> Calculate the speed of sound at 25 °C.

Speed of sound | Temperature dependence

Apply $v = 331 + 0.6T$ with $T = 25$ °C.

$v = 331 + 0.6(25) = 331 + 15$

$\boxed{v = 346 \text{ m/s}}$

Problem 2

> Light waves with $f = 7.0 \times 10^{14}$ Hz cannot be detected by most human eyes. Calculate the wavelength.

Wave equation | EM waves

Light travels at $c = 3.0 \times 10^8$ m/s. Use $\lambda = c/f$.

$$\lambda = \frac{3.0 \times 10^8}{7.0 \times 10^{14}} = 4.29 \times 10^{-7} \text{ m}$$

$\boxed{\lambda \approx 4.3 \times 10^{-7} \text{ m = 429 nm}}$

Note: 429 nm is actually visible (violet-blue). The problem statement may be slightly off — but the calculation is what matters.

Problem 3

> Wavelength in air at 20 °C of middle C (260 Hz).

Wave equation | Sound in air

$v = 331 + 0.6(20) = 343 \text{ m/s}$

$\lambda = v/f = 343/260$

$\boxed{\lambda \approx 1.32 \text{ m}}$

Problem 4

> Room 4.0 m × 5.2 m × 3.5 m. First 3 harmonics for each dimension. Any double/triple resonances?

Standing waves in rooms | Open-open pipe analog

A room dimension behaves like an open-open pipe: $f_n = nv/(2L)$, $v = 343$ m/s.

L	f_1 (Hz)	f_2 (Hz)	f_3 (Hz)
4.0 m	42.9	85.8	128.6
5.2 m	33.0	66.0	99.0
3.5 m	49.0	98.0	147.0

Coincidence check: 5.2 m's $f_3 \approx 99.0$ Hz vs. 3.5 m's $f_2 = 98.0$ Hz — these match within ~1% (≈ **double resonance** at ~98 Hz).

No triple resonance — no third dimension produces a frequency near 98 Hz.

Approximate double resonance at ≈ 98 Hz (5.2 m × 3.5 m).

Problem 5

> 1.5 W sound source, uniform in all directions. (a) Intensity at 2.5 m. (b) dB level. (c) dB level at 1.0 m.

Inverse-square law | Decibel scale

(a) $I = \dfrac{P}{4\pi r^2} = \dfrac{1.5}{4\pi(2.5)^2} = \dfrac{1.5}{78.54}$

$I_{2.5} \approx 1.91 \times 10^{-2}$ **W/m²**

(b) $\beta = 10\log_{10}\left(\dfrac{1.91 \times 10^{-2}}{1.0 \times 10^{-12}}\right) = 10\log_{10}(1.91 \times 10^{10})$

$\beta = 10(10.281) \approx \mathbf{102.8\ dB}$

(c) At $r = 1.0$ m: $I_{1.0} = \dfrac{1.5}{4\pi(1.0)^2} = 0.1194$ W/m²

$\beta = 10\log_{10}(1.194 \times 10^{11}) = 10(11.077) \approx \mathbf{110.8\ dB}$

★ *Pattern check:* moving from 2.5 m → 1.0 m (factor 2.5×) makes intensity $2.5^2 = 6.25\times$ larger, which is $10\log(6.25) = +8.0$ dB — and 110.8 − 102.8 = 8.0 dB ✓

(a) 1.91×10^{-2} **W/m²** **(b) 102.8 dB** **(c) 110.8 dB**

Problem 6

> Wave on string: $v = \sqrt{T/(m/L)}$. (a) Guitar B-string, $f_1 = 247$ Hz, $L = 60$ cm. (b) Rope $m = 160$ g, $L = 4.0$ m, $T = 400$ N.

String wave speed | Harmonic series (fixed-fixed)

(a) i. Fundamental: $\lambda_1 = 2L = 1.20$ m. Then $v = f_1\lambda_1 = 247 \times 1.20$.

$v = 296.4$ **m/s**

(a) ii. $f_3 = 3f_1 = 3(247)$.

$f_3 = 741$ **Hz**

(a) iii. $\mu = 0.01$ g/cm $= 1.0 \times 10^{-3}$ kg/m. From $v = \sqrt{T/\mu}$:
$T = \mu v^2 = (1.0 \times 10^{-3})(296.4)^2$.

$T \approx 87.9$ **N**

(b) $\mu = 0.160/4.0 = 0.040$ kg/m. $v = \sqrt{400/0.040} = \sqrt{10{,}000} = 100$ m/s.

(b) i. $\lambda_1 = 2L = 8.0$ m, $\lambda_2 = L = 4.0$ m.

(b) ii. $f_1 = v/\lambda_1 = 100/8 = 12.5$ Hz, $f_2 = 100/4 = 25$ Hz.

(a) 296.4 m/s · 741 Hz · 87.9 N | (b) λ: 8 m, 4 m f: 12.5 Hz, 25 Hz

Problem 7

String 0.85 m long. $f_3 = 780$ Hz. Find fundamental and propagation velocity.

Harmonic ratios

$f_n = nf_1 \Rightarrow f_1 = f_3/3 = 780/3 = 260$ Hz.

$\lambda_1 = 2L = 1.70$ m, $\; v = f_1\lambda_1 = 260 \times 1.70$.

$\boxed{f_1 = 260 \textbf{ Hz}, \quad v = 442 \textbf{ m/s}}$

Problem 8

Air column open at one end, $L = 0.075$ m. (a) Frequency at 20 °C. (b) % change at 50 °C.

Closed-end pipe ($f_n = nv/4L$) | Temperature effect

"Open at one end" → closed at the other. Fundamental: $f_1 = v/(4L)$.

At 20 °C: $v = 343$ m/s. $f_1 = 343/(4 \cdot 0.075) = 343/0.3$.

$f_1 = 1143$ **Hz**

At 50 °C: $v' = 331 + 0.6(50) = 361$ m/s. $f_1' = 361/0.3 = 1203$ Hz.

% change: $\dfrac{1203 - 1143}{1143} \times 100\%$.

$\boxed{f_1 \approx 1143 \textbf{ Hz}; \quad \approx +5.25\% \textbf{ at 50 °C}}$

Problem 9

50 cm organ pipe at 20 °C. First three harmonic frequencies for: (a) open both ends; (b) closed at one end.

Harmonic series · pipes

$L = 0.5$ m, $v = 343$ m/s.

(a) Open-open $\;f_n = nv/2L = n \cdot 343$ Hz.

$f_1 = 343$ Hz, $f_2 = 686$ Hz, $f_3 = 1029$ Hz

(b) Closed-one-end $\;f_n = nv/4L$, n odd. $v/4L = 171.5$ Hz.

$f_1 = 171.5$ Hz, $f_3 = 514.5$ Hz, $f_5 = 857.5$ Hz

(a) 343, 686, 1029 Hz | (b) 171.5, 514.5, 857.5 Hz

Problem 10

> Successive harmonics: 240 Hz, 280 Hz at 20 °C. (a) Open or closed? (b) Length?

Diagnostic — odd vs. all harmonics

Test "Open-open": successive harmonics differ by f_1. So $f_1 = 280 - 240 = 40$ Hz. Check: $240/40 = 6$, $280/40 = 7$ (consecutive integers ✓).

Test "Closed-one-end": only odd harmonics, so successive ones differ by $2f_1$. So $f_1 = 20$ Hz, but then $240/20 = 12$ — even, not allowed. ✗

(a) Open at both ends.

(b) $f_1 = v/(2L) = 40\text{ Hz} \Rightarrow L = v/(2 \cdot 40) = 343/80$.

(a) Open | (b) $L \approx 4.29$ m

Problem 11

> (a) Longitudinal vs. transverse. (b) Define: constructive/destructive interference, superposition, standing wave, node, antinode, harmonic, loudness, pitch, sound quality.

Definitions / vocabulary

(a) Longitudinal: medium oscillates *parallel* to wave propagation (compressions/rarefactions). E.g., sound. **Transverse:** medium oscillates *perpendicular* to wave propagation. E.g., string waves, light.

(b) Definitions:

- **Superposition** — when two waves overlap, the total displacement at every point is the algebraic sum of each individual displacement.
- **Constructive interference** — overlap in phase → amplitudes add (large displacement).
- **Destructive interference** — overlap out of phase (180°) → amplitudes subtract (zero or small displacement).
- **Standing wave** — pattern formed when two equal waves traveling in opposite directions interfere; produces fixed nodes and antinodes (no net energy transfer).
- **Node** — point on a standing wave with zero displacement (always still).
- **Antinode** — point with maximum displacement (oscillates with full amplitude).
- **Harmonic** — integer-multiple resonant frequency of a system: $f_n = nf_1$.
- **Loudness** — perception related to wave *intensity* (amplitude^2).
- **Pitch** — perception related to wave *frequency*.
- **Sound quality (timbre)** — perception related to the *mix of harmonics* present (waveform shape).

Vocab — see definitions above.

Problem 12

> French horn, $L = 3.7$ m, mouthpiece = closed end. First overtone is the lowest playable note. (a) Hand keeps far end open — frequency? (c) Hand closes far end — frequency?

Closed-open vs. closed-closed pipe | "First overtone" interpretation

$v = 343$ m/s. "First overtone" = next harmonic above the fundamental.

(a) Hand open → closed-one-end pipe. Allowed: $n = 1, 3, 5, \ldots$. The fundamental (f_1) is "not playable" → *first overtone* = f_3.

$$f_3 = \frac{3v}{4L} = \frac{3 \cdot 343}{4 \cdot 3.7} = \frac{1029}{14.8}.$$

$f_3 \approx 69.5$ **Hz**

(b) Sketches: displacement and pressure for $n = 1, 3, 5$ — closed end has displacement *node* (pressure antinode); open end has displacement *antinode* (pressure node). Pressure graph is the "complement" of the displacement graph.

(c) Hand closes → both ends closed. Both ends are displacement nodes. Allowed: $n = 1, 2, 3, \ldots$ with $f_n = nv/2L$. Fundamental not playable → *first overtone* = f_2.

$$f_1 = \frac{v}{2L} = \frac{343}{7.4} = 46.4 \text{ Hz}, \quad f_2 = 2f_1.$$

$f_2 \approx 92.7$ **Hz**

(d) Sketches: displacement nodes at *both* ends, antinodes in the middle ($n = 1$: 1 antinode in center; $n = 2$: 2 antinodes; $n = 3$: 3). Pressure graph: pressure antinodes at both ends, nodes between.

(a) ≈ 69.5 Hz | (c) ≈ 92.7 Hz

Problem 13

> Guitar string, $f_1 = 440$ Hz, $L = 0.85$ m. (a) Sketch first two modes. (b) i. Speed in string. ii. f_4. (c) Length needed for 660 Hz (same tension). (d) Crevasse 610 m, 13 s total time → temperature inside.

String harmonics | Free-fall + sound delay

(a) Sketches: 1st mode — single half-wavelength, nodes at both ends, one antinode in middle. 2nd mode — full wavelength, nodes at both ends and middle, two antinodes.

(b) i. $\lambda_1 = 2L = 1.70$ m, $\quad v = f_1\lambda_1 = 440 \times 1.70$.

$v = 748$ **m/s**

(b) ii. $f_4 = 4f_1 = 4(440)$.

$f_4 = 1760$ **Hz**

(c) Same string, same tension, same $\mu \to$ same $v = 748$ m/s. For new $f = 660$ Hz: $\lambda = v/f = 748/660 = 1.133$ m. Fundamental of new length: $L' = \lambda/2$.

$L' \approx 0.567$ **m (≈ 56.7 cm)**

(d) Total time = fall time + sound-travel time = 13 s.

Free-fall: $h = \frac{1}{2}g t_{\text{fall}}^2 \Rightarrow t_{\text{fall}} = \sqrt{2h/g} = \sqrt{2(610)/9.8} = \sqrt{124.5} \approx 11.16$ s.

Sound time: $t_{\text{sound}} = 13 - 11.16 = 1.84$ s. Speed of sound: $v = h/t_{\text{sound}} = 610/1.84 \approx 331.0$ m/s.

Solve for T: $\quad 331.0 = 331 + 0.6T \Rightarrow T \approx 0$ °C.

$T \approx 0$ **°C** *(makes sense — glaciers are at freezing!)*

(b) 748 m/s · 1760 Hz | (c) ≈ 0.567 m | (d) ≈ 0 °C

Problem 14

> 288 Hz tuning fork over water-filled glass tube. Resonances at $L = 0.28, 0.80, 1.40$ m. Glass tube total = 1.5 m. (a) Velocities. (b) Higher or lower freq for more resonances? (c) % diff vs. theoretical at 20 °C.

Resonance tube (closed-end) | Experimental error / end correction

(a) Closed-end pipe (water = closed end). Resonance lengths: $L_n = (2n-1)\lambda/4$, so $\lambda = 4L_n/(2n-1)$ and $v = f\lambda$.

n	L (m)	$\lambda = 4L/(2n-1)$	$v = 288\lambda$
1	0.28	1.120 m	**322.6 m/s**
2	0.80	1.067 m	**307.2 m/s**
3	1.40	1.120 m	**322.6 m/s**

★ *Note:* Resonance #2's discrepancy is due to **end correction** (effective tube length is slightly longer than measured). The two consistent values (#1 and #3) give $v \approx 322.6$ m/s.

(b) Higher frequency. Higher $f \to$ smaller $\lambda \to$ resonance positions $L_n = (2n-1)\lambda/4$ are *closer together*, so more resonances fit inside the 1.5 m tube.

(c) Theoretical at 20 °C: $v_{\text{theory}} = 343$ m/s. Experimental best estimate: $v_{\text{exp}} \approx 322.6$ m/s.

$$\% \text{ diff} = \frac{|343 - 322.6|}{343} \times 100\%.$$

≈ 5.95%

(a) 322.6, 307.2, 322.6 m/s | (b) Higher freq | (c) ≈ 6.0%